Exercise – Understanding the theory of block bootstrap

1. Aim of the exercise

The aim of this exercise is to understand how to correctly estimate the sampling distribution of a statistic (e.g., the sample mean) when data are serially correlated, such as in time-series settings. Standard bootstrap methods assume independent observations and can lead to biased inference. We use the block bootstrap method to account for dependence across observations and compare its performance to sampling directly from the population.

2. The block bootstrap

Sections 1 and 2 assume that the observations are i.i.d. In time series, however, observations that are close in time are typically correlated. This section shows that the bootstrap of Section 1 then fails, how the block bootstrap repairs this by resampling blocks of consecutive observations, and that the consistency of the block bootstrap requires conditions on the block length, a tuning parameter that the researcher has to choose[TK.

2.1. Serial dependence and the failure of the standard bootstrap

Let $X_1, \ldots, X_n$ be an ordered sample from a strictly stationary process with mean μ and autocovariances

$$\gamma_h = Cov(X_t, X_{t+h}), \quad h = 0, \pm 1, \pm 2, \ldots.$$

Stationarity means that the joint distribution of any stretch of consecutive observations does not change over time, so that γ_h depends only on the lag h. Because the observations are dependent, the marginal CDF F no longer fully describes how the sample is generated. We therefore consider the joint distribution of the process, denote it by P, and write the sampling CDF of the statistic T_n in (3) as $G_n(\tau, P)$.

Dependence changes the variance of T_n:

$$v_n = Var(T_n) = \sum_{h=-(n-1)}^{n-1} \left(1 - \frac{|h|}{n}\right) \gamma_h. \tag{1}$$

For i.i.d. data, only the term $h = 0$ remains and $v_n = \gamma_0 = \sigma^2$, as in Section 1. If the autocovariances are absolutely summable, v_n converges to the long-run variance

$$\sigma_\infty^2 = \sum_{h=-\infty}^{\infty} \gamma_h = \gamma_0 + 2\sum_{h=1}^{\infty} \gamma_h. \tag{2}$$

Under the weak-dependence conditions discussed in Section 3.3, a CLT for dependent data gives

$$G_\infty(\tau, P) = \Phi\left(\frac{\tau}{\sigma_\infty}\right).$$

With positive autocorrelation, $\sigma_\infty^2 > \gamma_0$: each observation partly repeats the information contained in previous observations, so the sample mean varies more than it would under independence.

The bootstrap of Section 1 draws observations independently from F_n and therefore destroys the dependence structure. Given the sample, the bootstrap observations are i.i.d. with mean $\bar{X}_n$ and variance

$$\gamma_0 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}_n\right)^2,$$

so that

$$Var^*(T_n^*) = \gamma_0 \xrightarrow{p} \gamma_0, \tag{3}$$

where $Var^*(\cdot)$ denotes variance with respect to the resampling distribution, holding the original sample fixed.

The bootstrap therefore approximates a distribution with variance γ_0 rather than σ_∞^2. Consequently, the consistency result (11) fails unless the autocovariances happen to cancel. As first noted by Singh1981, the standard bootstrap is generally inconsistent when observations are serially dependent.

2.2. The non-overlapping block bootstrap

The block bootstrap resamples blocks of consecutive observations instead of single observations, so that the dependence within each block is preserved. Choose a block length

$$l \in \{1, \ldots, n\},$$

and divide the sample into

$$k = \left\lfloor \frac{n}{l} \right\rfloor$$

non-overlapping blocks

$$Q_j = (X_{(j-1)l+1}, \ldots, X_{jl}), \qquad j = 1, \ldots, k. \tag{4}$$

If n is not a multiple of l, the last $n - kl$ observations are discarded, which leaves

$$m = kl$$

observations with mean $\bar{X}_m$.

A bootstrap sample

$$X_1^*, \ldots, X_m^*$$

is formed by drawing k blocks with replacement from $Q_1, \ldots, Q_k$ and placing them one after another. Observations within the same bootstrap block keep their original dependence, whereas observations in different blocks are independent.

Each block is drawn with probability $1/k$, so the bootstrap sample mean $\bar{X}_m^*$ has expectation $\bar{X}_m$, and the bootstrap counterpart of (3) is

$$T_n^* = \sqrt{m}\,(\bar{X}_m^* - \bar{X}_m). \tag{5}$$

Its CDF given the sample,

$$G_n(\tau, P_l),$$

where P_l denotes the distribution generated by resampling blocks of length l, takes the place of the bootstrap CDF (6) and is simulated with B bootstrap samples as in (8).

When $l = 1$, every block consists of a single observation, and the block bootstrap reduces to the standard bootstrap. The moving block bootstrap of Kunsch1989 uses all $n - l + 1$ overlapping blocks instead and is typically more accurate, although the role of the block length is the same Lahiri2003. We focus on the non-overlapping block bootstrap of Carlstein1986, because it makes the role of the block length most transparent.

2.3. The block length and the conditions for consistency

The role of the block length is clearest from the variance of T_n^*. Let $\bar{Q}_j$ denote the mean of block j. Because a bootstrap sample consists of k blocks drawn independently and with equal probability, $\bar{X}_m^*$ is the average of k independent draws from $\bar{Q}_1, \ldots, \bar{Q}_k$. The bootstrap variance of T_n^* is therefore

$$\sigma_l^2 = Var^*(T_n^*) = \frac{l}{k} \sum_{j=1}^{k} (\bar{Q}_j - \bar{X}_m)^2. \tag{6}$$

Each block is a stationary stretch of l observations, so

$$l\, Var(\bar{Q}_j) = v_l,$$

where v_l is defined as in (12), with l replacing n. Since $\bar{X}_m$ is the average of the block means, taking expectations in (17) yields

$$E(\sigma_l^2) = v_l - \frac{l}{m} v_m. \tag{7}$$

Equation (18) can be used to show why the block length must satisfy two conditions for consistency.

First, the block length must increase with the sample size:

$$l \to \infty.$$

The quantity v_l contains only the autocovariances up to lag $l - 1$, with weights $1 - h/l$, because the dependence between observations that end up in different blocks is broken when blocks are resampled independently. For a fixed block length, v_l therefore does not converge to σ_∞^2, and the bootstrap remains inconsistent no matter how large the sample becomes. When $l = 1$, (18) reduces to the expectation of γ_0 in (14). As l increases, v_l approaches σ_∞^2, but only gradually. If

$$\sum_{h=1}^{\infty} h\gamma_h < \infty,$$

then the bias satisfies

$$v_l - \sigma_\infty^2 \approx -\frac{2}{l} \sum_{h=1}^{\infty} h\gamma_h,$$

so that the bias is of order $1/l$.

Second, the number of blocks must also increase:

$$k \to \infty,$$

which is equivalent to

$$\frac{l}{n} \to 0.$$

Because the bootstrap variance in (17) is an average over k block means, its variance is of order

$$\frac{1}{k} = \frac{l}{n}.$$

If l grows proportionally to n, then k remains fixed, the final term in (18) does not vanish, and σ_l^2 remains random, no matter how large the sample becomes. When the blocks are long, their means are approximately independent and normally distributed, and σ_l^2 behaves approximately as

$$\sigma_\infty^2 \frac{\chi_{k-1}^2}{k}.$$

The choice of l therefore involves a bias–variance trade-off. Short blocks fail to capture enough dependence, producing a bias of order $1/l$. Long blocks leave too few blocks available for resampling, producing a variance of order l/n. Balancing the squared bias against the variance implies that the mean squared error of the bootstrap variance is minimized when

$$l \propto n^{1/3}.$$

Hall1995 show that the optimal rate depends on the purpose of the bootstrap: $l \propto n^{1/3}$ for estimating a bias or variance, $l \propto n^{1/4}$ for a one-sided distribution function, and $l \propto n^{1/5}$ for a two-sided distribution function. The constant of proportionality depends on the unknown autocovariances, so in practice the block length is typically chosen using a rule of thumb or a data-driven procedure PolitisWhite2004.

Formally, the block bootstrap for the sample mean is consistent if the process is strictly stationary, the dependence between observations decays sufficiently quickly as the lag increases (formalized through strong mixing conditions), a moment slightly higher than the second is finite, and the block length satisfies

$$l \to \infty \qquad \text{and} \qquad \frac{l}{n} \to 0.$$

Under these conditions Kunsch1989, Lahiri2003,

$$\sup_{\tau \in \mathbb{R}} |G_n(\tau, P_l) - G_n(\tau, P)| \xrightarrow{p} 0, \qquad n \to \infty. \tag{8}$$

Compared with (11), the statistical error now depends not only on the sample size but also on the block length chosen by the researcher, and it vanishes only if the block length grows with the sample size at a suitable rate. The simulation error is unaffected and continues to be controlled by the number of bootstrap replications B, as in (10).

2.4. Visualizing the convergence results

We demonstrate these conditions by simulation using a stationary Gaussian AR(1) process

$$X_t = \phi X_{t-1} + \varepsilon_t,$$

with $\phi = 0.7$ and i.i.d. innovations

$$\varepsilon_t \sim N(0, 1),$$

starting from its stationary distribution, so that $\mu = 0$. Its autocovariances are

$$\gamma_h = \frac{\phi^h}{1-\phi^2},$$

so that

$$\gamma_0 = 1.96,$$

whereas

$$\sigma_\infty^2 = \frac{1}{(1-\phi)^2} = 11.1.$$

The standard bootstrap variance (14) is therefore less than one fifth of the long-run variance. The dependence in this process decays geometrically, so the process satisfies the conditions discussed above. Because the process is Gaussian, $\bar{X}_n$ is exactly normally distributed and the sampling CDF is

$$G_n(\tau, P) = \Phi\left(\frac{\tau}{\sqrt{v_n}}\right),$$

with v_n given by (12). As in Section 2, the bootstrap can therefore be compared directly with the exact sampling distribution.

We compare three block-length rules:

$$l = 1,$$

the standard bootstrap, which violates the condition $l \to \infty$;

$$l = n^{1/3},$$

which satisfies both conditions and has the rate that is optimal for variance estimation; and

$$l = \frac{n}{5},$$

which violates the condition $l/n \to 0$ because it always uses

$$k = 5$$

blocks.

The sample sizes are

$$n = 125,\ 1000,\ 8000,$$

so that $n^{1/3}$ and $n/5$ are integers and no observations are discarded. As in Section 2, the samples consist of the first n observations of a single simulated sequence, and the number of bootstrap replications is fixed at

$$B = 10,000.$$

Figures 1 and 2 show the bootstrap PDFs and CDFs, with one row for each block-length rule and one column for each sample size. Because convergence in probability concerns behaviour across repeated samples, a single sample may provide a misleading impression. Figure 7 therefore shows the exact bootstrap variance (17), relative to the true variance v_n, across 1,000 independent samples. The MATLAB code is provided in Appendix B.

The three rows of Figures 1 and 2 correspond to the three cases distinguished in Section 3.3. With $l = 1$ (top row), the bootstrap distribution is much too narrow. Its variance is approximately 2.1 in every sample, close to $\gamma_0 = 1.96$, whereas the true variance is approximately 11. The largest distance between the bootstrap and sampling CDFs remains roughly constant at 0.19 for all three sample sizes (0.193, 0.190, and 0.197). Increasing the sample size therefore does not improve the approximation. The standard bootstrap converges, but it converges to the wrong distribution, exactly as predicted by (14).

With $l = n^{1/3}$ (middle row), the bootstrap distribution approaches the sampling distribution as the sample size increases. The largest distance between the bootstrap and sampling CDFs falls from 0.056 for $n = 125$ to 0.029 for $n = 1000$ and to 0.015 for $n = 8000$, consistent with the convergence result in (19).

With $l = n/5$ (bottom row), the bootstrap resamples only five blocks, so the bootstrap sample mean can take at most 126 different values, one for each possible draw of five blocks with replacement. Consequently, the estimated PDF is spiky and the estimated CDF is a step function. The largest distances (0.105, 0.261, and 0.082) do not decrease systematically with the sample size. In this case, the apparent accuracy of the bootstrap depends more on which five blocks happen to be observed than on the sample size itself.

Figure 3 shows that these patterns are systematic. For $l = 1$, the ratio of the bootstrap variance to the true variance concentrates more and more tightly around

$$\frac{\gamma_0}{\sigma_\infty^2} = 0.18.$$

Its mean is 0.176 when $n = 8000$, and its standard deviation decreases from 0.038 to 0.005. The bootstrap therefore becomes increasingly precise about the wrong value.

For $l = n^{1/3}$, the distribution shifts toward one and becomes more concentrated. The means are 0.51, 0.73, and 0.86, while the standard deviations are 0.17, 0.11, and 0.06. These means agree closely with the values 0.52, 0.73, and 0.86 implied by (18). Convergence is nevertheless slow. Even when $n = 8000$, the bootstrap still underestimates the variance by approximately 14% on average because the bias of order $1/l$ remains substantial when dependence is as persistent as in this process.

For $l = n/5$, the distribution changes little as the sample size increases. The mean remains between 0.69 and 0.80 and the standard deviation between 0.46 and 0.59, close to the values 0.80 and 0.57 implied by the approximation

$$\sigma_\infty^2 \frac{\chi_4^2}{5}.$$

With only a fixed number of blocks, a larger sample does not make the bootstrap more accurate.

The simulation therefore illustrates the trade-off faced by the researcher when choosing the block length. Blocks that are too short capture too little dependence and yield a bootstrap distribution that is precise but systematically incorrect. Blocks that are too long leave too few blocks for resampling and yield a bootstrap distribution that remains noisy regardless of the sample size. Even when the block length grows at the theoretically optimal rate, a large sample may still be required before the bootstrap approximation becomes accurate.

```
%% 3. Construct the data-generating process and analytical distributions

% 3.1. Set a seed for the random number generator for reproducibility
```

```
rng(26)

% 3.2. Set the autoregressive parameter of the AR(1) process
phi = 0.7;

% 3.3. Set the standard deviation of the innovations
sigma_eps = 1;

% 3.4. Compute the variance of the AR(1) process
gamma_0 = sigma_eps^2 / (1 - phi^2);

% 3.5. Define a function that simulates a stationary AR(1) series from a
% vector of innovations. The first observation is drawn from the
% stationary distribution, so that the series is stationary from the
% start.
ar1 = @(epsilon) filter(1,[1 -phi], ...
    [epsilon(1) / sqrt(1 - phi^2); epsilon(2:end)]);

% 3.6. Define the variance of the statistic for sample size n
v_pop = @(n) gamma_0 * (1 + 2 * sum((1 - (1:n-1) / n) .* phi.^(1:n-1)));

% 3.7. Define the sampling PDF of the statistic
g_pop = @(tau,n) pdf('Normal',tau,0,sqrt(v_pop(n)));

% 3.8. Define the sampling CDF of the statistic
G_pop = @(tau,n) cdf('Normal',tau,0,sqrt(v_pop(n)));

% 3.9. Set the grid of tau values at which PDFs and CDFs are evaluated
tau_grid = linspace(-15,15,1000)';

% 3.10. Set the sample sizes
N_obs_vec = [125 1000 8000];

% 3.11. Draw a sequence of observations from the AR(1) process
data_all = ar1(random('Normal',0,sigma_eps,[max(N_obs_vec),1]));

% The statistic of interest is Tn = sqrt(n) * (sample mean - mu), with mu =
% 0. Because the AR(1) process is Gaussian, the sample mean is normally
% distributed, and the sampling distribution of Tn is known analytically:
% Tn ~ N(0, v_n). This allows the bootstrap approximation to be compared
% directly with the true sampling distribution. The sample sizes are chosen
% so that n^(1/3) and n/5 are integers.

%% 4. Set the block-length rules

% 4.1. Define the block-length rules as functions of the sample size
block_rule = {@(n) 1, @(n) round(n^(1/3)), @(n) n / 5};

% 4.2. Set the labels of the block-length rules
block_label = {'l = 1', 'l = n^{1/3}', 'l = n/5'};
```

```
54
55  % 4.3. Set the number of block-length rules
56  N_rules = length(block_rule);
57
58  % 4.4. Set the sample sizes
59  N_n = length(N_obs_vec);
60
61  %% 5. Construct block bootstrap distributions for different n and l
62
63  % 5.1. Set the number of bootstrap replications
64  N_boot = 10^4;
65
66  % 5.2. Preallocate storage for the bootstrap statistic
67  T_boot = NaN(N_boot,N_rules,N_n);
68
69  % 5.3. For each block-length rule and sample size, select the original
70  % sample, divide it into non-overlapping blocks, draw bootstrap samples of
71  % blocks, and compute the bootstrap statistic
72  for j = 1:N_rules
73      for i = 1:N_n
74          n = N_obs_vec(i);
75          l = block_rule{j}(n);
76          k = n / l;
77          data_sample = data_all(1:n);
78          % Each row of blocks is one block of l consecutive observations
79          blocks = reshape(data_sample,l,k)';
80          for b = 1:N_boot
81              data_boot = datasample(blocks,k,1,'Replace',true);
82              % Tn* = sqrt(n) * (bootstrap sample mean -
83              % original sample mean)
84              T_boot(b,j,i) = sqrt(n) * ...
85                  (mean(data_boot(:)) - mean(data_sample));
86          end
87      end
88  end
89
90  %% 6. Convergence of the block bootstrap PDF to the sampling PDF
91
92  % Plot the block bootstrap PDF and the true sampling PDF for each
93  % block-length rule (rows) and sample size (columns).
94  figure('Position',[100 100 1000 1000]);
95  layout = tiledlayout(N_rules,N_n, ...
96      'TileSpacing','compact', ...
97      'Padding','compact');
98  for j = 1:N_rules
99      for i = 1:N_n
100         n = N_obs_vec(i);
101         l = block_rule{j}(n);
102         nexttile
103         hold on
```

```
        plot(tau_grid, ...
            ksdensity(T_boot(:,j,i),tau_grid,'function','pdf'),'r')
        plot(tau_grid, ...
            g_pop(tau_grid,n),'k--')
        title(['n = ' num2str(n) ',l = ' num2str(l)])
        xlim([tau_grid(1) tau_grid(end)])
        if i == 1
            ylabel(block_label{j},'FontWeight','bold')
        end
        hold off
    end
end
xlabel(layout,'\tau')
ylabel(layout,'Density')
lgd = legend( ...
    'Block bootstrap PDF', ...
    'Sampling PDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout,['Fig. 1. Block bootstrap PDF and sampling PDF ' ...
    'as n grows (B = ' num2str(N_boot) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 7. Convergence of the block bootstrap CDF to the sampling CDF

% Plot the block bootstrap CDF and the true sampling CDF for each
% block-length rule (rows) and sample size (columns). The title of each
% panel reports the largest distance between the two CDFs over the grid of
% tau values.
dist_max = NaN(N_rules,N_n);
figure('Position',[100 100 1000 1000]);
layout = tiledlayout(N_rules,N_n, ...
    'TileSpacing','compact', ...
    'Padding','compact');
for j = 1:N_rules
    for i = 1:N_n
        n = N_obs_vec(i);
        l = block_rule{j}(n);
        G_boot = arrayfun(@(t) mean(T_boot(:,j,i) <= t),tau_grid);
        dist_max(j,i) = max(abs(G_boot - G_pop(tau_grid,n)));
        nexttile
        hold on
        plot(tau_grid,G_boot,'r')
        plot(tau_grid,G_pop(tau_grid,n),'k--')
        title({['n = ' num2str(n) ',l = ' num2str(l)], ...
            ['Largest distance: ' num2str(dist_max(j,i),'%.3f')]})
        xlim([tau_grid(1) tau_grid(end)])
        if i == 1
            ylabel(block_label{j},'FontWeight','bold')
```

```
        end
        hold off
    end
end
xlabel(layout,'\tau')
ylabel(layout,'Cumulative probability')
lgd = legend( ...
    'Block bootstrap CDF', ...
    'Sampling CDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout,['Fig. 2. Block bootstrap CDF and sampling CDF ' ...
    'as n grows (B = ' num2str(N_boot) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 8. Block bootstrap variance across repeated samples

% Figures 5 and 6 are based on a single sample for each n. To see whether
% the block bootstrap converges in probability, compute the bootstrap
% variance of Tn* in many independent samples. Because all blocks have the
% same length, the bootstrap variance equals l times the mean squared
% deviation of the block means from the sample mean, so it can be computed
% exactly without drawing bootstrap samples.

% 8.1. Set the number of repeated samples
N_rep = 1000;

% 8.2. Preallocate storage for the ratio of the bootstrap variance to the
% true variance of Tn
var_ratio = NaN(N_rep,N_rules,N_n);

% 8.3. For each repeated sample and sample size, draw a new AR(1) series
% and compute the bootstrap variance for each block-length rule
for r = 1:N_rep
    for i = 1:N_n
        n = N_obs_vec(i);
        data_rep = ar1(random('Normal',0,sigma_eps,[n,1]));
        for j = 1:N_rules
            l = block_rule{j}(n);
            k = n / l;
            block_means = mean(reshape(data_rep,l,k),1)';
            var_boot = l * mean((block_means - mean(data_rep)).^2);
            var_ratio(r,j,i) = var_boot / v_pop(n);
        end
    end
end

% 8.4. Set the grid of ratio values at which the PDFs are evaluated
ratio_grid = linspace(0,3,3001)';
```

```
204
205 % 8.5. Plot the PDF of the variance ratio across repeated samples for each
206 % block-length rule and sample size. A ratio of 1 means that the bootstrap
207 % variance equals the true variance.
208 figure('Position',[100 100 1200 450]);
209 layout = tiledlayout(1,N_rules, ...
210     'TileSpacing','compact', ...
211     'Padding','compact');
212 for j = 1:N_rules
213     nexttile
214     hold on
215     for i = 1:N_n
216         plot(ratio_grid,ksdensity(var_ratio(:,j,i),ratio_grid, ...
217             'function','pdf','Support','positive', ...
218             'BoundaryCorrection','reflection'))
219     end
220     xline(1,'k--');
221     title(block_label{j})
222     xlim([ratio_grid(1) ratio_grid(end)])
223     hold off
224 end
225 xlabel(layout,'Bootstrap variance / true variance')
226 ylabel(layout,'Density')
227 lgd = legend( ...
228     ['n = ' num2str(N_obs_vec(1))], ...
229     ['n = ' num2str(N_obs_vec(2))], ...
230     ['n = ' num2str(N_obs_vec(3))], ...
231     'True variance', ...
232     'Orientation','horizontal');
233 lgd.Layout.Tile = 'south';
234 t = title(layout,['Fig. 3. Block bootstrap variance across ' ...
235     num2str(N_rep) ' samples']);
236 t.FontSize = get(groot,'defaultAxesFontSize');
237 t.FontWeight = 'normal';
238
239 %% 9. Summarize the results
240
241 % Print, for each block-length rule and sample size, the block length, the
242 % number of blocks, the true variance of Tn, the bootstrap variance in the
243 % single sample of Figures 5 and 6, the largest distance between the CDFs,
244 % and the mean and standard deviation of the variance ratio across the
245 % repeated samples.
246 fprintf('\n%-12s %6s %6s %6s %8s %8s %8s %8s %8s\n','Rule', ...
247     'n','l','k','v_n','Var*','Dist','MeanRat','SDRat')
248 for j = 1:N_rules
249     for i = 1:N_n
250         n = N_obs_vec(i);
251         l = block_rule{j}(n);
252         fprintf('%-12s %6d %6d %6d %8.3f %8.3f %8.3f %8.3f %8.3f\n', ...
253             block_label{j},n,l,n/l,v_pop(n),var(T_boot(:,j,i)), ...
```

```
            dist_max(j,i),mean(var_ratio(:,j,i)),std(var_ratio(:,j,i)))
    end
end
fprintf('Variance of the AR(1) process (gamma_0): %.3f\n',gamma_0)
fprintf('Long-run variance sigma_eps^2/(1-phi)^2: %.3f\n', ...
    sigma_eps^2 / (1 - phi)^2)
```

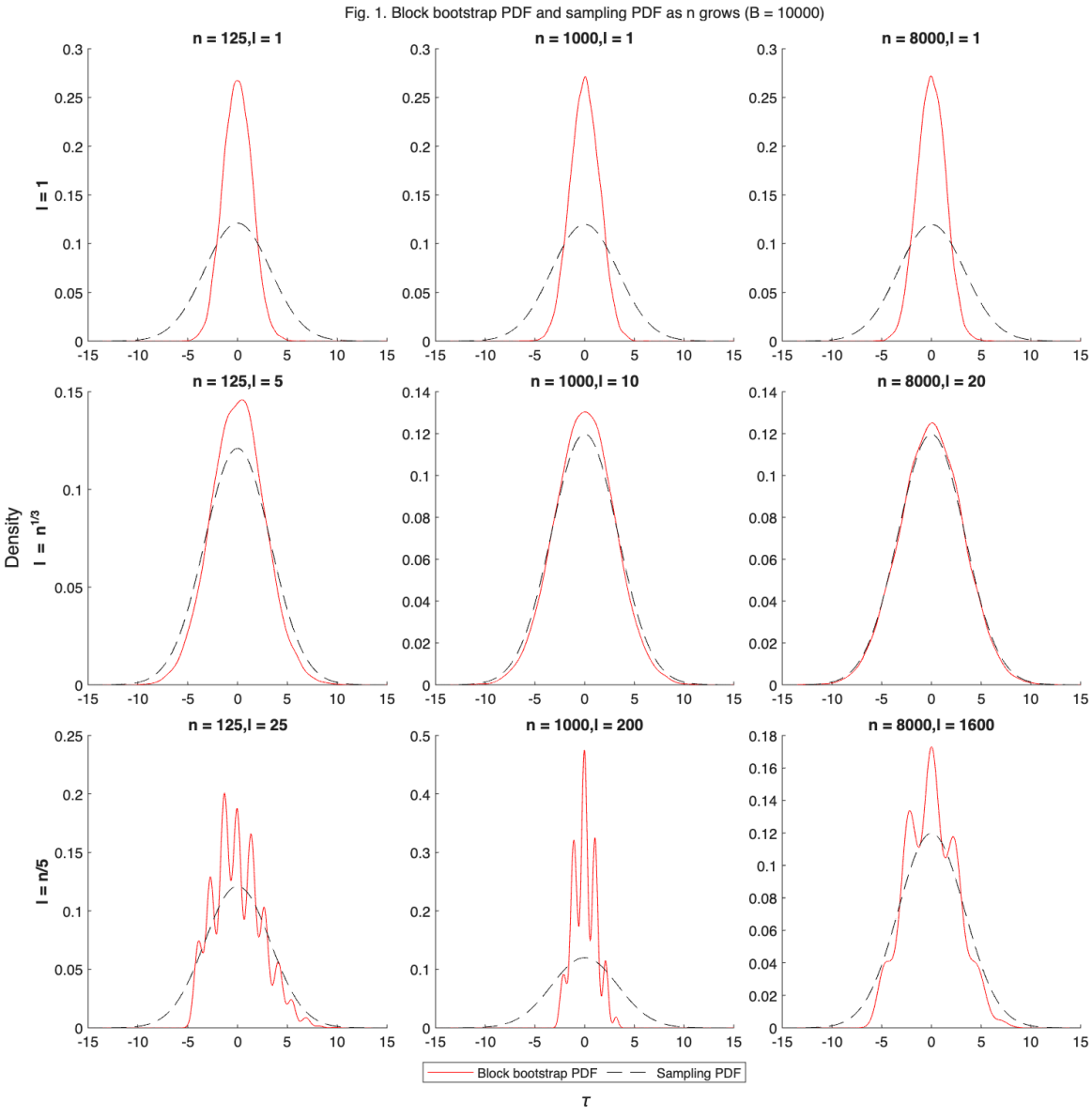

Fig. 1. Block bootstrap PDF and sampling PDF as n grows (B = 10000)
n = 125,l = 1
n = 1000,l = 1
n = 8000,l = 1
n = 125,l = 5
n = 1000,l = 10
n = 8000,l = 20
n = 125,l = 25
n = 1000,l = 200
n = 8000,l = 1600
l = 1
l = n^1/3
l = n/5
Density
Block bootstrap PDF
Sampling PDF
τ

Fig. 2. Block bootstrap CDF and sampling CDF as n grows (B = 10000)

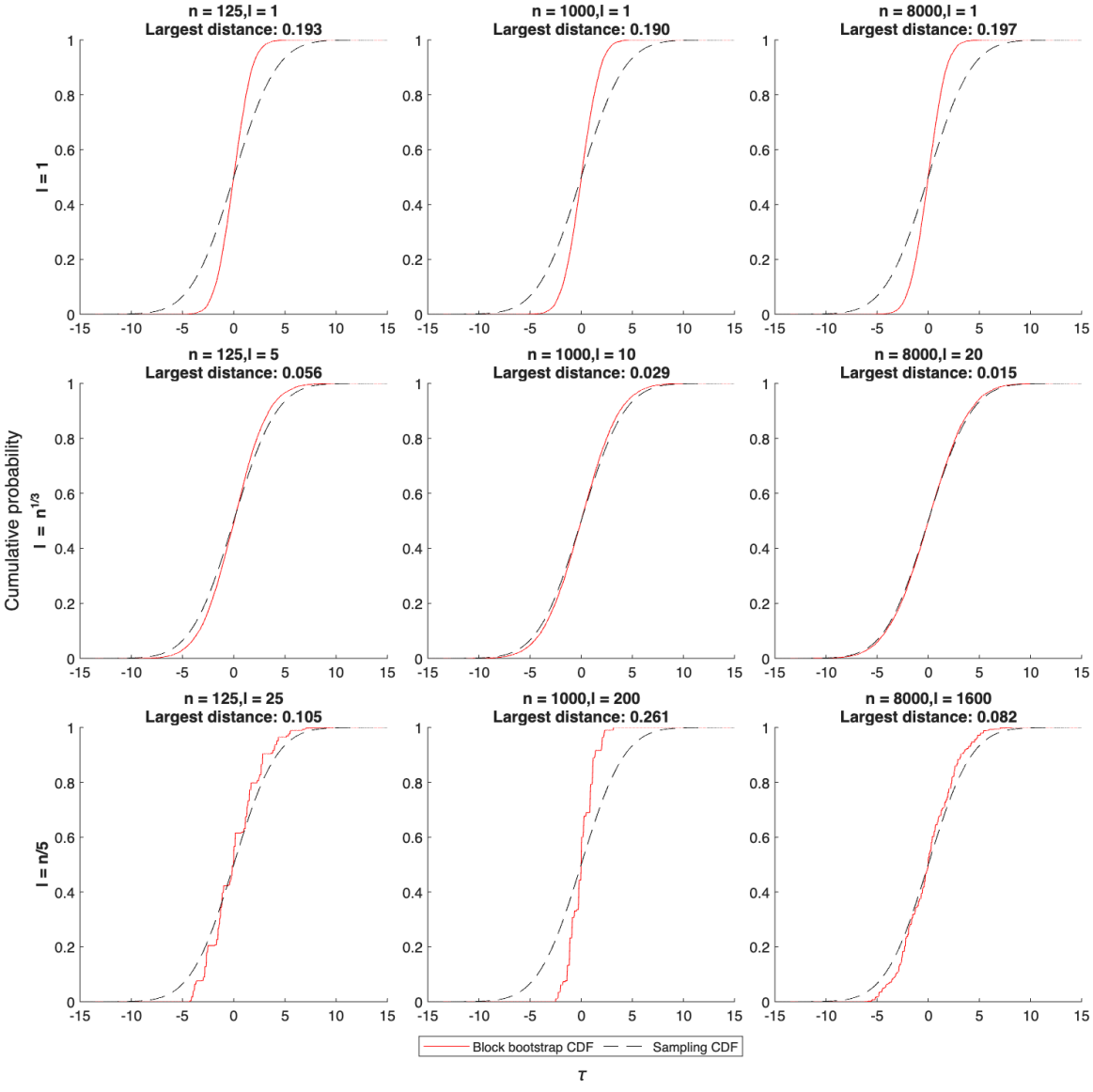
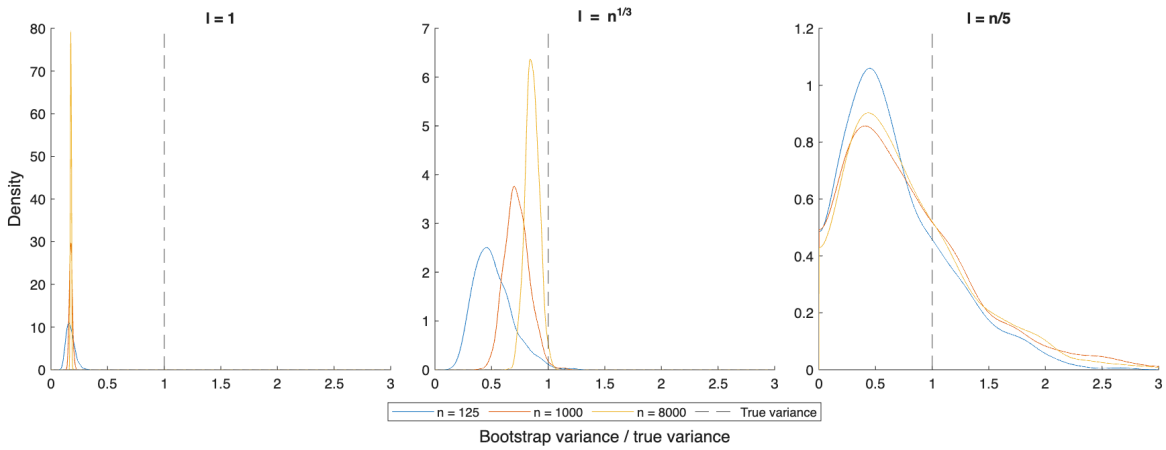


Fig. 3. Block bootstrap variance across 1000 samples



3. Final notes

This file is prepared and copyrighted by Noortje Kamphuis, Axel Zoons and Tunga Kantarcı. This file and the accompanying MATLAB file are available on GitHub and can be accessed using this link. This file has made use of four references. First is Radovanov and Marcikić, 2014. A comparison of four different block bootstrap methods. Croatian Operational Research Review. Second is Mignani and Rosa, 1995). The moving block bootstrap to assess the accuracy of statistical estimates in Ising model simulations. Computer physics communications. The third is Nordman and Lahiri, 2014. Convergence rates of empirical block length selectors for block bootstrap. The fourth is Hall, P., Horowitz, J. L., Jing, B. Y., 1995. On blocking rules for the bootstrap with dependent data. Biometrika, 82(3), 561-574.